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The Repeated Binomial Model and the Binomial Distribution in Excel

I. CONSTRUCTING THE BINOMIAL TREE AND FINAL OUTCOMES

Consider the so-called "stationary recombining tree": there are $N+1$ dates, $k \in \{0, 1, ..., N-1, N\}$. At each date (except the last one), the an asset with price S_k can go up to $S_{k+1} = uS_k$ or down to dS_k where $u > 1 + r > d$, for (net) risk-free rate r.

1. Make a sample tree in Excel with $N = 3$, $u = 4/3$, $d = 3/4$, and $S_0 = 144$. In the first column you should have just S_0, the Date-0 price, in the second column you should have uS_0 and dS_0, the possible Date-1 prices, and so on: in the $(k+1)$st column, you should have uS_k and dS_k for each value of S_k in the nth column.

   Note that this tree is recombining: different paths can lead to the same final realization. For example, you can get to $S_3 = S_0u^2d$ by going up at dates 0 and 1, and then going down at date 2 or, alternatively, by going down at date 0 and then going up at

dates 1 and 2. But keep all paths in your Excel file: at each Date k, you should have 2^k rows.

What are all the possible realizations of S_N?

2. In general, the possible realizations are $\{u^k d^{N-k} S_0 \,;\, k \in \{0,...,N\}\}$. Try to explain this formula intuitively.

   Construct this set of possible realizations in Excel for the example above. I.e. for $N = 3$, populate a column with $k = 0,...,N$ and then for each k find $S_N = 144 \times (4/3)^k (3/4)^{N-k}$.

3. Compare your answer in question 2 to your answer in question 1.

   Show that the different methods give the same possible outcomes.

II. The Distribution of Final Outcomes

1. Consider the outcome $S_N = S_0 u^k d^{N-k}$ for each $k \in \{0,...,3\}$. How many rows do you have in your example that given this number?
2. What is the $\mathbb{Q}$-probability of an up move at any node? Call this number q.
3. What is the $\mathbb{Q}$-probability of each of the final rows in your tree in question I.1 above?

   Populate a column with these numbers.
4. Combine your answers in questions 3 and 1 above to populate a column with the $\mathbb{Q}$-probabilities of each final outcome. I.e. for each $k \in \{0,...,3\}$ you should have a cell with the $\mathbb{Q}\left[S_N = S_0 u^k d^{N-k}\right]$.
5. For $k \in \{0,...,3\}$ use the `BINOM.DIST()` function to populate a column with the binomial probabilities of k "successes" (or "ups" here) for number of trials $N = 3$ and the probability of success q you found above.
6. Compare your answer in question 5 to your answer in question 4.

   Show that the numbers you found in question 4 are binomial probabilities.

We have shown that the number of ups is binomially distributed. However, we don't know how variable $S_N = S_0 u^k d^{N-k}$ is distributed yet, because the number of ups is in the exponent. We'll figure out how do deal with this below.

III. The Binomial Distribution and the Normal Distribution

Now consider large N, e.g., $N = 50$.

1. Populate three columns:

a) the number of ups $k = 0, \ldots, N$;

b) the binomial probability of each number of ups from the `BINOM.DIST()` function;

c) the cumulative binomial distribution of the number of ups from the `BINOM.DIST()` function.

Start by doing this with your the probability q you found above. I will ask you to do it for different probabilities, so you probably will want to refer to a cell with this probability in it.

2. Create graphs for your answers in question 1. I.e. create a graphs with k on the horizontal axis and the values of the `BINOM.DIST()` function on the vertical axis.
3. Using the columns you created above, calculate the mean and variance of the number of ups using the formula $\mathbb{E}[k] = \sum_{k=0}^{N} k\mathbb{Q}[k]$ and $\mathrm{Var}[k] = \sum_{k=0}^{N} (k - \mathbb{E}[k])^2$.
4. Using your answer above, show that show that $\mathbb{E}[k] = qN$ and $\mathrm{Var}[k] = q(1-q)N$. Do this for a number of different values of q, e.g., $q \in \{.1, .5, .7\}$.
5. Denote $\mathbb{E}[k]$ by μ and $\mathrm{Var}[k]$ by σ^2.

   Using the functions `PI()`, `EXP()`, and `SQRT()`, populate a column with the number $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(k-\mu)^2}{2\sigma^2}}$ for each k.
6. Compare your answer to question 5 with your answer from question 1 (ii) above. (You may want to look up the De Moivre–Laplace theorem.)
7. Viewing your answer to question 5 as a probability density, calculate the associated cumulative distribution function. I.e. create a column in which each cell is the sum of all the values in the rows above it in the preceding column.
8. Use the `NORM.DIST()` function to populate a column with the cumulative normal distribution function with mean μ and variance σ. I.e. the column should contain the probability that the number of ups is less than k for each k.
9. Compare your answer to question with your answer from question 1c above.

IV. The Distribution of S_N

We have now shown that the number of ups is approximately normally distributed. Now we investigate how $S_N = S_0 u^k d^{N-k}$ distributed.

1. Use the laws of logarithms (basically $\log ab = \log a + \log b$) to show that

$$\log S_N = \log S_0 + N \log d + k \log\left(\frac{u}{d}\right)$$

2. If k is normally distributed with mean μ and variance σ^2, how is $\log\left(\frac{S_N}{S_0}\right)$ distributed?

We have now shown that in the with many steps, asset prices on recombining binomial tree are approximately log-normally distributed under $\mathbb{Q}$ (i.e. the log of S_T is normal). Given this distribution, we can calculate the prices of many derivatives directly from the FTAP, $X_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[X_T]$. Indeed, just this kind of calculation will lead to the famous Black–Scholes formula.